a bit clearer

C PROGRAMMING FROM SCRATCH · EPISODE 01

Introduction to C

Concepts and History

KEY POINTS

- A program is a precise set of instructions; C is one language for writing them.
- BCPL → B → C, and C grew up together with Unix.
- Ritchie created C; Kernighan co-wrote the book (K&R).
- 1989 ANSI C89, 1990 ISO C90. A standard is a shared rulebook, not a guarantee.

01 What Is C?

▶ VIDEO 00:44

People talk to each other in **natural languages**. To tell a computer what to do, we write a **program**, a precise set of instructions, in a **programming language**. C is one of many, alongside C++, Java, Go and Python.

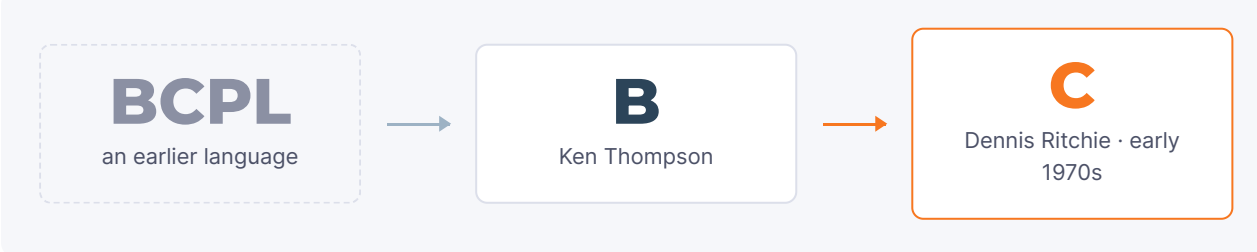
> KEY IDEA
>
> **C is a language for describing tasks to a computer. We write a program in C; the computer carries out the task by following it.**

02 Where C Came From

▶ VIDEO 01:57

YEAR	MILESTONE
1969	At Bell Labs, Ken Thompson, Dennis Ritchie and colleagues build early **Unix**, written in assembly language (low-level, tied to one machine). Thompson designs **B**, based on **BCPL**.
1972	In the early 1970s Ritchie builds on B to create **C**. 1972 is the key year. The name simply takes the next letter.
1973	The Unix **kernel**, the core of the operating system, is rewritten in C. As Unix spreads, C spreads with it.
1978	*The C Programming Language* is published, known as **K&R** after its two authors.

FIGURE 1.1 **From BCPL to B to C**



All along the way, the stories of C and Unix were tightly woven together.

WATCH OUT **Keep the two names straight**

Dennis Ritchie was the main creator of C. **Brian Kernighan** co-authored the book with him but didn't create the language.

03 Why Standards Matter

▶ VIDEO 04:37

As C spread, different compilers implemented it in slightly different ways and added their own extensions. A program could compile in one environment and need changes in another. A **language standard** fixes this: a shared rulebook for compiler developers and programmers.

YEAR	EVENT	NAME
1989	ANSI (American National Standards Institute) approves a standard for C	**C89**, also "ANSI C"
1990	Adopted as an international standard, ISO/IEC 9899:1990	**C90**

KEY IDEA

A standard gives a common foundation for moving programs between environments. It does **not** guarantee that every C program runs unchanged everywhere.

04 C Today

▶ VIDEO 06:21

More than fifty years on (here in 2026), C is still widely used in operating systems and other system software, and the TIOBE index of language popularity has ranked it among the leading languages over the years. Dennis Ritchie passed away in 2011; C and Unix continue to shape computing today.